DO NOW, AO1 REVISION

Stopping distance

Forces
Paper 2, Combined and Separate
14 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**What is the stopping distance of a vehicle?**	Thinking distance + braking distance	☐
2 ★	**What is the thinking distance of a vehicle?**	The distance travelled during the driver's reaction time.	☐
3 ★	**What is the braking distance of a vehicle?**	The distance travelled under the braking force.	☐
4	**Name the factor that increases both thinking distance and braking distance.**	Speed	☐
5	**State the typical range of human reaction times.**	0.2 s to 0.9 s	☐
6 ★	**Give four factors that affect reaction time.**	Tiredness, drugs, alcohol and distractions	☐
7 ★	**Give four factors that affect braking distance.**	Speed, wet or icy roads, worn tyres and worn brakes	☐
8 ★	**Describe the energy transfer when a vehicle brakes.**	Energy is transferred from the kinetic store of the vehicle to the thermal store of the brakes.	☐
9	**Describe how the braking force needed to stop in a given distance depends on speed.**	The greater the speed, the greater the braking force needed.	☐
10 ★	**Give two dangers of a very large deceleration.**	The brakes may overheat and the driver may lose control.	☐
11	**Describe a simple method to measure a person's reaction time.**	Drop a ruler between their fingers. The further it falls before they catch it, the longer the reaction time.	☐
12	**Explain why braking distance increases with speed.**	The vehicle has more kinetic energy, so the brakes must do more work to stop it.	☐

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	QUESTION	ANSWER	✓
13	**The speed of a vehicle doubles. State what happens to the thinking distance.**	It doubles (it is directly proportional to speed).	☐
14	**Which does speed affect more: thinking distance or braking distance?**	Braking distance	☐